

HETERO Program Manual

BenchMarking Process

This is the file path:

```
hetero--tmt-main > src > main > java > com > hetero > test >
```

1. Select a Data Structure to Evaluate

```
=====
      COMPLEXITY ANALYSER & BENCHMARK TOOL
=====
Select a Data Structure to evaluate:
1. ArrayList (ArrayListTaskRepo)
2. LinkedList (LinkedListTaskRepo)
3. HashMap (HashMapTaskRepo)
4. Run All Structures (Side-by-Side Comparison)
5. Exit Program
Enter your choice (1-5):
```

2. In this example, we used HashMap

```
Enter your choice (1-5): 3
Enter the number of requests/tasks to simulate: █
```

3. You can directly enter the number of tasks to simulate. We enter 125 here

```
Enter the number of requests/tasks to simulate: 125 █
```

4. The Output is easily seen as the full analysis of each function

STRUCTURE PROFILE: HASHMAPTASKREPO		
Feature Operation Evaluated	Execution (Time)	Heap Space (Delta)
1. Add/Insert Task (Bulk Setup)	1.5599 ms	121.29 KB
2. List All Tasks (findAll)	0.0636 ms	60.73 KB
3. Update Task Details (update)	0.0117 ms	0.00 KB
4. Find Task by ID (findById)	0.0148 ms	60.23 KB
5. Filter by Completion Status	0.0960 ms	60.26 KB
6. Filter by Tasks Due Today	0.0943 ms	60.23 KB
7. Filter by Category	0.0723 ms	60.26 KB
8. Filter by Priority	0.0568 ms	60.26 KB
9. Delete Task (delete)	0.0112 ms	0.00 KB

The steps are applicable for all three data structures.